KRITICAL HACKS 2026
AI / ML TRACK

HEARTLINE

SMART AMBULANCE ROUTING & TRAFFIC ALERT SYSTEM

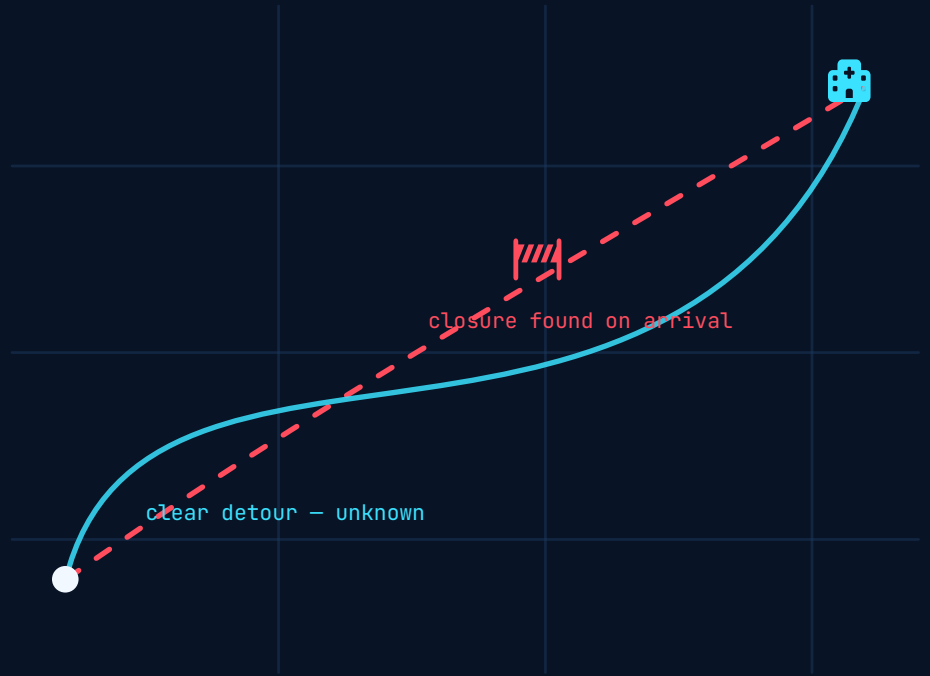
Smarter routes. Earlier alerts. Safer passage.



5ive Minds
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IGDTUW

Minutes lost in traffic decide outcomes.

Three failures stack up on every emergency run — and everyone on the road pays for them.



Today: the direct corridor is blocked, a clear detour exists — but nothing tells the crew in time.

Congestion delays the response

Ambulance crews rely on experience and radio chatter. Jams and closures often surface only when the vehicle is already stuck in them.

Road disruptions stay invisible

Roadworks, accidents and waterlogging are published as advisories scattered across official pages — ingested late, or never, by navigation tools.

Nearby drivers can't yield early

Drivers first learn about an approaching ambulance when they hear the siren — too late to calmly clear a lane.

WHO CARRIES THE COST

- Patients & families
- Ambulance crews
- Nearby drivers
- Hospitals & dispatch

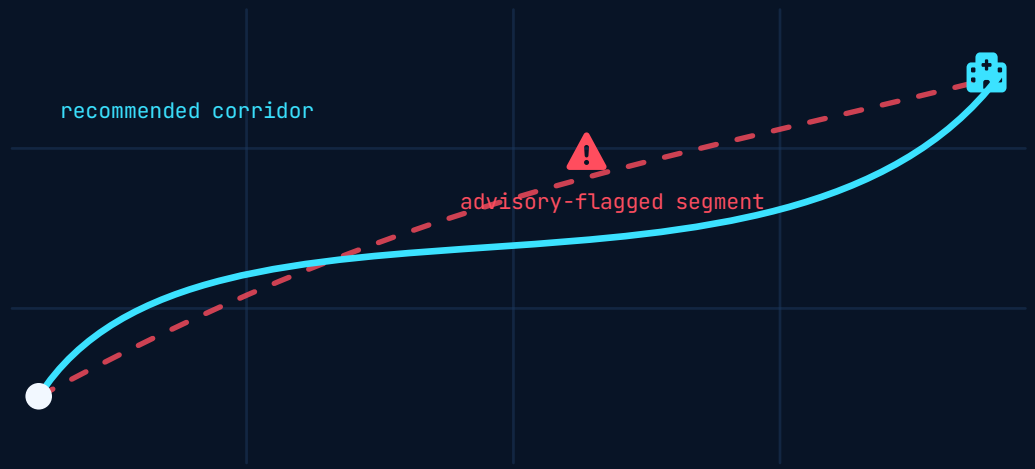
Two features. One live corridor.

HeartLine pairs smarter routing for the ambulance with earlier alerts for opted-in drivers on its path.

01 Smart route recommendation

FOR THE AMBULANCE CREW

The crew enters the origin and destination. HeartLine fetches candidate routes, checks recorded demo obstacles, and recommends a route based on obstacle counts and distance.



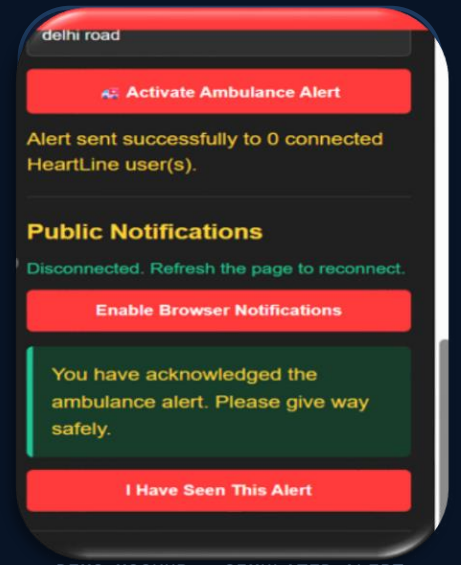
02 Opt-in corridor alerts

FOR NEARBY DRIVERS

When the operator activates an ambulance alert, connected HeartLine users can receive it through the browser-based prototype. Users can acknowledge the alert; browser notifications require permission.

Design principle

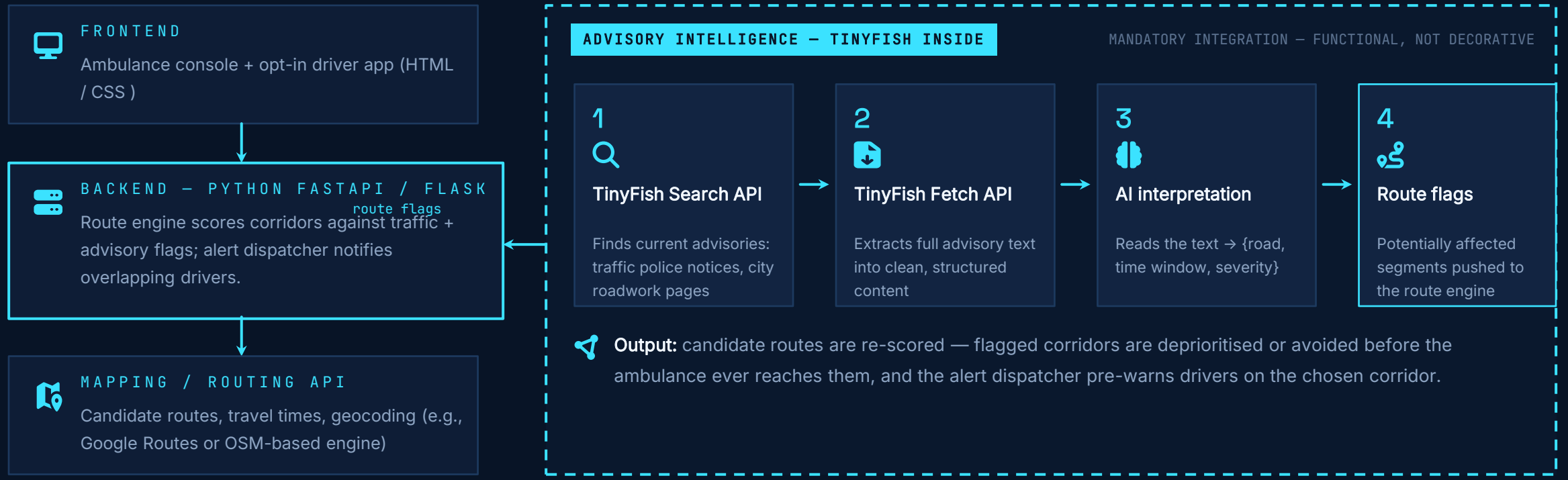
Alerts are opt-in and session-based. No persistent location tracking



DEMO MOCKUP — SIMULATED ALERT

Advisories become route intelligence.

TinyFish is the advisory intake layer — it finds and extracts the road notices that routing APIs don't carry.



Why TinyFish is load-bearing: official advisories live on scattered web pages that routing APIs don't ingest. Search discovers them, Fetch extracts them, and the AI layer turns prose into machine-readable signals the router can act on.

Built to demo honestly.

This is the actual proposed stack — and the status labels say exactly what the demo does and doesn't do.

PROPOSED STACK

Frontend — HTML · CSS · JavaScript
Map view, crew console, driver alert screen

Backend — Python, Flask / FastAPI
Route scoring, advisory parsing, alert dispatch logic

Mapping API — Leaflet + OpenStreetMap + Nominatim + OSRM
Map display, location lookup, route calculation and alternative-route selection

TinyFish — Search API + Fetch API
Advisory discovery and extraction — mandatory integration

AI interpretation — LLM of choice
Interpret retrieved advisories and classify potential route disruptions, once an LLM is integrated

Notifications —WebSocket + Browser Notifications API
Broadcasts ambulance alerts to connected users and displays browser notifications when permission is granted

PROTOTYPE WORKFLOW

1. **Enter origin & destination** — the crew sets the run in the web console.
2. **Fetch candidate routes** — the backend calls the mapping API for options and travel times.
3. **Retrieve demo obstacles**— Search + Fetch feed the AI layer; affected segments get flagged.
4. **Show the recommendation** — the console displays the recommended route, while the driver alert is simulated in the browser.

● IMPLEMENTED — route request + advisory flag demo

○ SIMULATED — live traffic & driver alerts, fixed scenario

○ NOT YET BUILT — fleet integration, push delivery

At pilot scale, running costs stay near zero.

Verified prices separated from estimates — anything unconfirmed is marked for vendor confirmation, not guessed.

COMPONENT	WHAT IT COVERS	COST BASIS	MONTHLY ESTIMATE
Backend hosting	Flask / FastAPI app + worker	Free tier or small cloud VM — roughly $0–10 / month	₹0–₹850
Mapping & routing	Candidate routes, geocoding	Leaflet map, OpenStreetMap tiles, Nominatim geocoding and OSRM routing	₹0–₹500 contingency
TinyFish Search + Fetch	Advisory discovery + extraction	Free within published usage limits; confirm limits before deployment	₹0 within published limits
AI interpretation	Advisories → route flags	LLM API, per-call pricing — model-dependent; small at advisory volume	₹0–₹1,000, if integrated
Notifications	Driver alerts	displayed only when the user grants notification permission	₹0

COST STRATEGY

- Use free tiers and open-source mapping tools where practical.
- Start with simulated alerts and limited testing.
- Monitor API usage before scaling.
- Recalculate costs for live traffic data and ambulance fleet deployment.

A subscription hypothesis — priced from evidence, not pitch math.

Who pays, why they'd pay, what stands in the way — and how we validate before we promise anything.

WHO PAYS

Ambulance fleet operators

Primary customers — private and contracted emergency fleets

Hospital networks

Facilities running their own patient transport

City emergency dispatch

Longer-term public-sector path, via tenders

REVENUE MODEL — HYPOTHESIS

TO BE VALIDATED

B2B subscription: per-vehicle monthly fee, tiered by fleet size, with a pilot tier for evaluation. The price range is deliberately unset — it will be derived from pilot measurements and willingness-to-pay interviews, not invented for this deck.

Customer value

Fewer surprise blockages on active runs, advisory-aware routing crews can trust, and a documented alert trail for every trip — sellable as reliability, not novelty.

Adoption challenges

Long public procurement cycles; integration with existing dispatch (CAD) systems; liability questions when a recommendation is wrong; driver opt-in rates still unknown. We'd rather name these than hide them.

Pricing validation plan

1) Pilot with one fleet — log reroute frequency and ETA variance. 2) Willingness-to-pay interviews with fleet managers. 3) Set the price from measured value per run, then revisit quarterly.

Expected impact if validated: Better visibility into reported road disruptions, more informed route recommendations, and earlier alerts for ambulance crews. Improvements in ETA reliability and response time will be measured during pilot testing.

Market-size figures are intentionally omitted — we have no credible primary research yet, and we won't present numbers we can't defend.

From demo corridor to city-scale safe passage.

Everything below is planned work — none of it is claimed as built.

WHAT COMES NEXT

Authorised fleet integration

Connect to licensed ambulance dispatch systems, not just a demo console

Reliable live traffic data

Partner-grade traffic feeds with latency guarantees for time-critical routing

Privacy-protective alerts at scale

Coarse geohash location, session-only sharing, automatic deletion

Pilot deployments

One hospital network + one fleet, with measured outcomes before expansion

Expansion opportunities

Fire and police corridors, more cities, V2X alerts as vehicles gain support

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THANK YOU

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